

StudyVault – AI-Powered Note Organizer

Description

StudyVault is a cross-platform mobile application that helps students centralize, organize, and understand their academic materials by managing PDFs and images in a single platform. It addresses the common problem of study resources being scattered across WhatsApp, Google Drive and local device storage, making them difficult to locate and review, especially during examinations.

Method

The application was developed using React Native (Expo) and TypeScript for the frontend, with Firebase Authentication for user management and Cloud Firestore for storing note and folder metadata. A hybrid storage architecture was adopted, where uploaded files are stored locally on the device while Firestore maintains structured metadata, reducing cloud storage costs and improving file access performance. The application includes note organization through folders, pinning, relevance-ranked search, light/dark themes and full CRUD functionality for notes and folders.

AI contribution

StudyVault integrates Google Gemini 2.5 Flash to provide intelligent study assistance. Uploaded PDFs and images are processed directly by Gemini to generate structured summaries containing key concepts, important topics, formulas and worked examples. These summaries are then reused to generate AI-powered revision quizzes without reprocessing the original document, providing a faster and more cost-effective study experience.

Novelty

Unlike conventional storage platforms that only manage files or note-taking applications that lack intelligent document understanding, StudyVault combines structured file organization with AI-powered summarization and quiz generation within a single lightweight mobile application. Its hybrid storage architecture and AI-assisted revision workflow provide students with both efficient material management and rapid exam preparation in one integrated solution.

GitHub repository:

<https://github.com/anushka-rodrigo/Study-Vault.git>

Demo video:

https://drive.google.com/file/d/1Zlj6I0uLUvQSHV6fhE7xWcB9Wb1ntrX/view?usp=drive_link